Portfolio Optimization using Markowitz Theory and the Excel Solver

Course Name: Portfolio Theory and Risk Management
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1. Introduction

Portfolio optimization is a fundamental concept in modern finance that focuses on selecting the best combination of assets to maximize return while minimizing risk. This project applies the Markowitz Mean-Variance Optimization framework to construct an optimal portfolio of selected U.S. stocks using Excel Solver.

The main objective is to analyze how diversification can reduce risk and improve portfolio performance. By combining theoretical concepts with practical implementation in Excel, this project demonstrates how investors can make informed decisions based on quantitative analysis.

2. Data Description

Five U.S. stocks were selected for analysis:

- Apple Inc. (**AAPL**)
- Microsoft Corporation (**MSFT**)
- Procter & Gamble Company (**PG**)
- Johnson & Johnson (**JNJ**)
- JPMorgan Chase & Co. (**JPM**)

Daily closing price data over approximately two months were collected and used to compute daily returns.

The risk-free rate used is:

- Annual: 4.303%
- Daily: 0.017075%

3. Methodology

3.1 Portfolio Return

The expected return of the portfolio is calculated as:

$$\boldsymbol{E(R_p) = \sum w_i r_i}$$

where $w_i$represents the weight of each stock and $r_i$represents its expected return.

3.2 Portfolio Risk (Standard Deviation)

Portfolio risk is measured using standard deviation, calculated as:

$$\sigma_p = \sqrt{W^T \Sigma W}$$

where:

- W= vector of weights
- Σ= covariance matrix

3.3 Sharpe Ratio

The Sharpe ratio measures risk-adjusted return:

$$\textbf{Sharpe Ratio} = \frac{E(R_p) - r_f}{\sigma_p}$$

where r_f is the risk-free rate.

3.4 Optimization using Excel Solver

Excel Solver was used to find:

- Maximum Return Portfolio
- Minimum Variance Portfolio
- Maximum Sharpe Ratio Portfolio
- Efficient Frontier (by varying target returns)

Constraints applied:

- Sum of weights = 1
- No short selling (weights ≥ 0)

4. Results and Findings

4.1 Individual Stock Statistics:

Stock	Expected Return (%)	Std Dev (%)	Variance	Sharpe Ratio
AAPL	-0.02147%	1.69819%	0.00028839	-0.02270
MSFT	-0.34830%	1.71984%	0.00029579	-0.20252
PG	-0.11510%	1.31408%	0.00017268	-0.08759
JNJ	0.17535%	0.98204%	0.00009644	0.17856
JPM	-0.07017%	1.63304%	0.00026668	-0.04297

Interpretation of Results:

The results indicate significant differences in performance and risk across the selected stocks.

- **Johnson & Johnson (JNJ)** is the only stock with a positive expected return (0.17535%) and has the lowest standard deviation (0.98204%), indicating it is the least risky asset in the sample. It also has the highest Sharpe ratio (0.178561), suggesting superior risk-adjusted performance.
- **Microsoft Corporation (MSFT)** shows the worst performance, with a highly negative expected return (-0.34830%) and the lowest Sharpe ratio (-0.20252), indicating poor risk-adjusted returns.
- **Apple Inc. (AAPL)** and **JPMorgan Chase & Co. (JPM)** exhibit moderate levels of risk but slightly negative returns, resulting in negative Sharpe ratios.
- **Procter & Gamble Company (PG)** has relatively lower volatility compared to most stocks, but its negative return leads to a negative Sharpe ratio.

4.2 Portfolio Optimization Results:

- **Minimum Variance Portfolio:**

The minimum variance portfolio represents the portfolio with the lowest possible risk (standard deviation) among all feasible portfolios. This portfolio is obtained by optimizing the asset weights using Excel Solver while minimizing portfolio variance subject to the constraint that the sum of weights equals one.

	AAPL	MSFT	PG	JNJ	JPM
Min variance weights	7.17%	24.62%	16.25%	47.93%	4.03%
$E(r_p)$	-0.0248%	/day	-6.055%	/year	
σ_p	0.6713%	$(w\Sigma w^T)^{1/2}$			
$E(r_p) - r_f/\sigma_p$	**(0.062355)**				

Interpretation:

- The portfolio achieves the **lowest possible risk (0.6713%)**, making it highly stable.
- The heavy allocation to low-risk stocks demonstrates the importance of **defensive assets in risk reduction**.
- Despite low risk, the portfolio produces a **negative expected return**, highlighting the trade-off between risk and return.
- The positive Sharpe ratio (0.0624) indicates that the portfolio still slightly outperforms the risk-free rate on a risk-adjusted basis.

The results show that the minimum variance portfolio places a significant weight on Johnson & Johnson (47.93%), which is consistent with its relatively low volatility and stable performance observed earlier. Defensive stocks such as Procter & Gamble Company and Microsoft Corporation also receive moderate allocations.

In contrast, more volatile or lower-performing assets such as Apple Inc. and JPMorgan Chase & Co. receive smaller weights.

- **Maximum Sharpe Ratio Portfolio:**

The maximum Sharpe ratio portfolio represents the optimal portfolio that maximizes risk-adjusted return. This portfolio is obtained by maximizing the ratio of excess return to risk using Excel Solver.

	AAPL	MSFT	PG	JNJ	JPM
Maximize $E(r_p)/\sigma_p$	0.00%	0.00%	0.00%	100.00%	0.00%
$E(r_p)$	0.1754%	/day	55.505%	/year	
σ_p	0.9820%	$(w\Sigma w^T)^{1/2}$			
$E(r_p) - r_f/\sigma_p$	**0.161174**				

Interpretation:

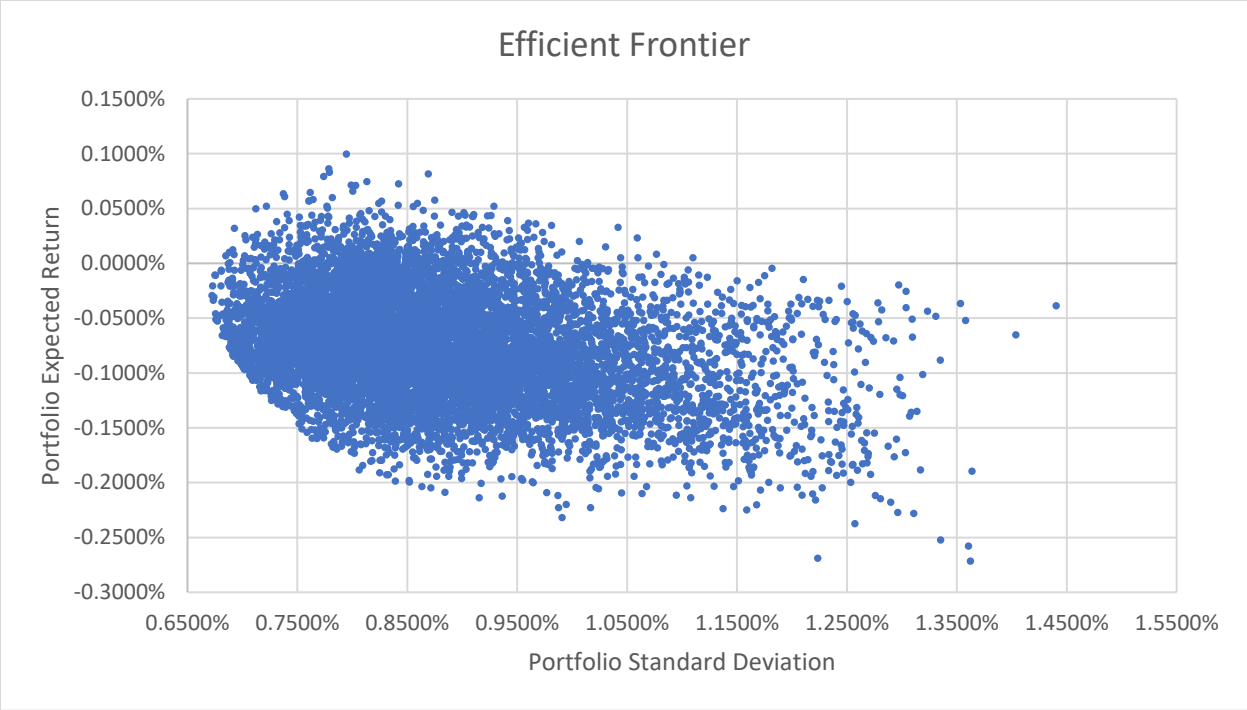
- JNJ dominates the portfolio because it has:
  - Positive expected return
  - Lowest volatility
  - Highest Sharpe ratio among all stocks
- All other stocks receive zero weight, indicating they do not improve the portfolio's risk-return trade-off.

The results indicate that the optimal portfolio allocates **100% of the investment to Johnson & Johnson (JNJ)**. This occurs because JNJ provides the **highest risk-adjusted return (Sharpe ratio)** among all available assets.

Unlike the minimum variance portfolio, which focuses on minimizing risk through diversification, the maximum Sharpe ratio portfolio selects the asset that offers the best balance between return and volatility.

4.3 Efficient Frontier

The efficient frontier represents the set of optimal portfolios that provide the highest expected return for a given level of risk. It is constructed by plotting portfolio standard deviation (risk) on the horizontal axis and expected return on the vertical axis.



Interpretation:

- The **upper boundary of the cloud** represents an **efficient frontier**.
- Portfolios on this boundary are **optimal**, meaning no other portfolio offers higher return for the same level of risk.
- Portfolios below this boundary are **inefficient**, as investors can achieve better returns without increasing risk.
- Most portfolios lie in the region with negative returns, reflecting the poor performance of several individual stocks during the sample period.
- The leftmost part of the graph corresponds to the minimum variance portfolio, which has the lowest risk.
- As we move to the right:
    - Risk increases
    - Potential return also increases
- The efficient frontier shows a positive risk-return trade-off, which is consistent with financial theory.
- This graph validates the Markowitz theory that diversification allows investors to construct portfolios that dominate inefficient combinations of assets.

5. Critical Analysis of Markowitz Theory

Although the Markowitz model is powerful, it has several limitations:

5.1 Assumption of Normal Distribution:

The model assumes that returns are normally distributed, which is not always true in real markets. Financial returns often exhibit skewness and kurtosis.

5.2 Sensitivity to Input Data:

Small changes in expected returns or covariance can significantly affect the optimal portfolio. This makes the model unstable in practice.

5.3 Historical Data Limitation:

The model relies on past data to predict future performance, which may not always be accurate.

5.4 Ignoring Transaction Costs:

The model does not consider transaction costs, taxes, or market liquidity, which are important in real-world investment.

5.5 No Short Selling Constraint:

In this project, short selling was not allowed, which may limit potential optimization outcomes.

6. Problems Encountered

During the implementation, several issues were encountered:

- Incorrect Sharpe ratio calculation due to improper use of the risk-free rate
- Errors in standard deviation due to incorrect data ranges
- Difficulty in setting up Solver constraints properly
- Axis formatting issues in the efficient frontier graph
- Initial misunderstanding of covariance matrix calculation

These issues were resolved by carefully reviewing formulas and ensuring consistency in data.

7. Conclusion

This project successfully demonstrates how Markowitz portfolio theory can be applied to construct optimal portfolios using Excel. The analysis highlights the importance of diversification and the trade-off between risk and return.

The maximum Sharpe ratio portfolio provides the most efficient allocation of assets, while the efficient frontier visually represents optimal investment choices.

While the Markowitz model offers a structured approach to portfolio optimization, its reliance on simplifying assumptions and sensitivity to input data highlights the need for cautious interpretation and integration with real-world considerations.

Despite its limitations, the Markowitz model remains a foundational tool in portfolio management and provides valuable insights for investors.

8. References

- Course materials and Excel template
- Financial data sources (Excel STOCKHISTORY)
- Assignment instructions
- YouTube. Portfolio Optimization using Markowitz Theory [Video].
  Available at: https://www.youtube.com/watch?v=IRa64LEERhE